

Dynamic-Neighborhood Quality-Guided World Models for Online Multi-Car Overtaking in Autonomous Racing

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Abstract—Online multi-car overtaking requires the interaction representation to change as nearby vehicles become more or less relevant. This paper presents Dynamic-Neighborhood Quality-Guided Deep Latent Competition (DNQ-DLC), a world-model planning framework that combines online graph construction with quality-guided prediction and control. At each real control step, DNQ-DLC scores map-conditioned traffic relations, reconstructs an ego-centric interaction graph, and encodes the selected vehicles in a bounded mask-padded tensor. This design keeps the neural interface fixed while the represented vehicle identities and traffic density change. A Graph-DLC world model predicts short-horizon packed states, reward, boundary risk, and uncertainty. A receding-horizon objective then selects overtaking actions by balancing progress, interaction quality, and track adherence. On a 200-case matched simulator benchmark, DNQ-DLC achieved an 85.5% overtaking-success rate, compared with 71.5% for Rule Expert, corresponding to a paired increase of 14.0 percentage points. These results support a pass-completion advantage for the complete controller in the evaluated simulator. Component tests do not show that every module is individually superior. Code, data, checkpoints, figures, and simulation videos are available at <https://chocho2222.github.io/DNQ-DLC/>.

Index Terms—Autonomous racing, overtaking, world models, dynamic interaction graphs, multi-agent decision making, receding-horizon planning.

I. Introduction

AUTONOMOUS racing is a demanding testbed for intelligent transportation systems. Learning-based agents have reached champion-level performance in head-to-head simulated car racing [1], showing the need for fast closed-loop decision making near vehicle limits. Autonomous-driving evaluation also faces a scarcity of safety-critical interactions. Naturalistic and adversarial testing has therefore become important for exposing rare failures [2], [3].

Overtaking concentrates these challenges into a short time window. The ego vehicle must find a passing opportunity, act on it, and remain within the track corridor as nearby competitors change position. The most relevant opponent can change from a leading vehicle to a lateral or rear vehicle within one maneuver. Online overtaking is therefore both a control problem and an interaction-selection problem.

Full-stack racing systems have demonstrated high-speed time-trial and oval operation [4], [5]. Head-to-head and multi-vehicle studies extend this setting to changing opponent behavior [6], [7]. Tube-based control provides a fur-

ther model-based approach to multi-vehicle racing [8]. A survey of autonomous racing summarizes the main system-level challenges [9]. Reviews of learned driving discuss its broader strengths and limitations [10], [11]. Many learned controllers, however, still use a fixed observation layout or require retraining when the number and arrangement of surrounding vehicles change. This is restrictive in multi-car overtaking because the interaction topology changes online.

World models allow a controller to compare short-horizon futures before executing an action. This idea supports general latent control [12], model-based driving [13], [14], and Deep Latent Competition (DLC) for racing [15]. A fixed vehicle topology still limits such models. The planner may retain irrelevant agents or omit the vehicle that currently constrains the pass. A progress-only objective can also favor fast but off-track maneuvers.

This paper proposes DNQ-DLC, a dynamic-neighborhood quality-guided world-model framework for online multi-car overtaking. At every decision step, the framework projects traffic states into a map-conditioned representation, reconstructs the interaction graph, and selects the most relevant surrounding vehicles. The variable physical neighborhood is converted into a bounded graph tensor and processed by a Graph-DLC world model for short-horizon prediction. Quality-guided action evaluation then converts predicted state, reward, boundary risk, and uncertainty into an overtaking decision. The physical interaction set can therefore change online without changing the neural input dimension or binding representation slots to fixed vehicle identities.

The main technical contributions are:

- We propose an online dynamic-neighborhood graph construction mechanism for multi-car racing. It rebuilds map-conditioned interactions from current vehicle states and represents changing vehicle identities and counts through a bounded, permutation-stable graph tensor.
- We develop a dynamic-graph Graph-DLC world model with quality-guided short-horizon planning. The model jointly predicts future state, reward, boundary risk, and uncertainty, enabling overtaking actions to be evaluated as interaction relationships evolve.
- We conduct extensive closed-loop experiments against rule-based, DLC, and multi-seed reinforcement-

learning baselines. On 200 matched cases, DNQ-DLC achieves an 85.5% overtaking-success rate, exceeding Rule Expert by 14.0 percentage points. Held-out and randomized evaluations provide complementary validation across multiple tracks and scenarios with 4–8 vehicles.

The remainder of this paper is organized as follows. Section II reviews autonomous racing, learned policies, world models, and interaction-aware representations. Section III presents the dynamic-neighborhood interface, Graph-DLC world model, and quality-guided planner. Section IV describes the evaluation protocol and experimental results. Section V discusses the supported claims and their boundaries, and Section VI concludes the paper and outlines future real-vehicle validation.

II. Related Work

Autonomous racing has evolved into a high-speed full-stack benchmark for intelligent transportation systems. The survey by Betz et al. [9] characterizes autonomous vehicle racing as a domain that simultaneously stresses perception, planning, control, vehicle dynamics, and system integration. System-level studies such as the TUM autonomous motorsport stack and the KAIST IAC system further show that head-to-head racing requires reliable real-time decision making under changing opponent behavior [4], [6]. More recent A2RL work demonstrates that autonomous racing continues to move toward full-scale multi-agent operation at the limits of vehicle handling [16]. These racing systems motivate the setting of this paper, but they also reveal a central difficulty: dense multi-car overtaking requires the planner not only to control the ego vehicle, but also to decide which surrounding vehicles should influence the next decision.

Optimization-based racing planners provide one way to address this problem when the interaction structure is known or can be specified compactly. Multi-vehicle racing planners and tube-MPC formulations have shown strong performance in high-speed racing scenarios [7], [8]. Their strength lies in systematic optimization and constraint handling, but their performance depends on the quality of the assumed vehicle model, opponent model, and interaction set. In dense overtaking, the relevant opponents can change within a few decision steps: a forward vehicle may become irrelevant after the ego vehicle initiates a pass, while a lateral or rear vehicle may become the dominant constraint. This motivates a representation that selects the interaction set online rather than keeping it fixed.

Learned policies provide another route to adaptive racing behavior. Deep reinforcement learning has demonstrated highly competitive racing performance in high-fidelity game environments [1], and recent F1TENTH work shows that overtaking behaviors can be learned and transferred to a physical small-scale racing platform [17]. Surveys of deep reinforcement learning for autonomous driving highlight the potential of learned policies for

decision making under complex dynamics, but also point out their sensitivity to state representation, training distribution, and reward design [10]. This limitation is especially important for online multi-car overtaking. A policy trained with predetermined observation semantics may implicitly associate a particular input slot with a particular opponent role, even though the physical vehicle occupying that role may change during the maneuver. DNQ-DLC therefore separates proposal generation from interaction selection: learned actors consume a bounded mask-padded tensorization, while the dynamic-neighborhood module rebuilds the physical interaction graph and determines which vehicles are represented at the current decision step.

The limitations of direct policy execution motivate predictive planners that can compare candidate actions before committing to one. World models learn dynamics that support planning in imagined futures [12]. In autonomous driving, MILE connects model-based imitation learning with closed-loop urban driving [13], while GAIA-1 demonstrates that generative world models can represent complex driving-scene evolution [14]. Recent driving world-model surveys summarize this direction as a promising way to model scene evolution and support interaction with dynamic environments [18]. More recent planning-oriented world-model methods further emphasize that learned dynamics should be aligned with downstream planning objectives: AdaWM studies adaptive world-model planning under distribution shift [19], WorldRFT uses reinforcement fine-tuning to align latent world-model representations with safety-critical planning behavior [20], and VLA-World combines predictive imagination with reflective reasoning for autonomous-driving foresight [21].

These world-model studies show the value of prediction for decision making, but autonomous racing imposes a sharper requirement. The planner must decide whether an overtaking attempt is desirable under short horizons, narrow lateral margins, and close interactions. A rollout that maximizes progress alone may still produce off-track exposure or unsafe side-by-side behavior. Therefore, DNQ-DLC uses the world model for candidate action evaluation rather than passive forecasting. Candidate rollouts are ranked by overtaking progress, track adherence, lateral quality, close-gap risk, predicted boundary risk, rollout uncertainty, and consistency with learned actor priors.

For world-model planning to be effective in traffic, the model must receive an interaction representation that remains valid as the number and relevance of nearby agents change. A recent survey of autonomous vehicle motion forecasting emphasizes that predicting the future behavior of surrounding traffic participants is essential for downstream decision making [22]. Large-scale datasets such as Waymo Open Motion and Argoverse 2 provide benchmarks for interactive multi-agent prediction [23], [24]. Transformer-based forecasting models such as Scene Transformer and Wayformer show that attention-based interaction modeling can capture dependencies among multiple agents [25], [26]. Map-relative and vectorized representations also play an important role: DenseTNT

formulates trajectory prediction through dense goal sets, and HiVT uses hierarchical vectorized interaction modeling for multi-agent motion prediction [27], [28].

Most of these interaction models target forecasting or urban traffic prediction rather than online racing control. Related game-theoretic and traffic-generation methods further emphasize the importance of explicit multi-agent relations, as seen in GameFormer and TrafficGen [29], [30]. DNQ-DLC adapts these interaction-modeling ideas to a receding-horizon racing controller. At each decision step, the method rebuilds a map-relative ego-centered graph, tensorizes the currently relevant vehicles into a bounded masked observation, and uses this runtime graph representation inside a world-model planner. Thus, the graph is not only a forecasting input; it determines which vehicles are imagined, scored, and acted upon in the next overtaking decision.

III. Methodology

This section presents DNQ-DLC, a dynamic-neighborhood quality-guided world-model planner for online multi-car overtaking in autonomous racing. DNQ-DLC operates as a receding-horizon decision layer above the low-level vehicle dynamics controller. At each decision step, it constructs a map-conditioned dynamic-neighborhood graph, packs the selected interaction-relevant vehicles into a bounded masked graph tensor, predicts short-horizon futures under candidate controls, and executes the first action of the highest-scoring rollout.

The method contains three main computational stages after problem formulation: dynamic map-conditioned neighborhood encoding, Graph-DLC world modeling, and quality-guided receding-horizon planning. The packed-observation layout, self-feature definition, network architecture, and numerical configuration are detailed in Supplementary Section I. Training labels, losses, and baseline budgets are reported in Supplementary Section II. Candidate generation and scoring are specified in Supplementary Section III, and metric definitions and experimental protocols are defined in Supplementary Section IV.

Notation convention. Subscript t denotes the physical decision time. Superscript q denotes an arbitrary vehicle when defining local geometric quantities, superscript j denotes a surrounding vehicle in an ego-neighbor relation, superscript i denotes an agent-centered row, superscript k denotes a packed relation slot, and superscript (n) denotes a candidate rollout. A hatted variable denotes an imagined quantity predicted by the world model. The notation $t+k|t$ denotes a k -step prediction made from decision time t . Variables without row superscript refer to the controlled ego row; for example, $z_t \equiv z_t^0$ and $h_t \equiv h_t^0$.

A. Problem Formulation

At decision time t , the ego vehicle observes a track map $\mathcal{M}$, its own state x_t^0 , and a set of surrounding vehicles

$$\mathcal{I}_t = \{1, 2, \dots, N_t^{\text{veh}}\}. \quad (1)$$

The ego state includes position, velocity, heading, track-relative progress, lateral deviation, and driving-status indicators. Surrounding vehicles are represented by the same physical quantities when available. State normalization is specified in Supplementary Section I-B and Table SII. The frozen numerical configuration is listed in Supplementary Section I-G and Table SIV.

The controller outputs a normalized low-level racing action

$$a_t = [\delta_t, u_t, b_t]^\top \in \mathcal{A}, \quad (2)$$

where δ_t , u_t , and b_t denote steering, throttle, and brake. The admissible action set is

$$\mathcal{A} = [\delta_{\min}, \delta_{\max}] \times [u_{\min}, u_{\max}] \times [b_{\min}, b_{\max}]. \quad (3)$$

For each candidate first action $a_t^{(n)}$, DNQ-DLC forms a short-horizon action continuation

$$\mathbf{A}_t^{(n)} = (a_t^{(n)}, \hat{a}_{t+1|t}^{(n)}, \dots, \hat{a}_{t+H-1|t}^{(n)}), \quad (4)$$

evaluates the imagined future with a multi-objective score $J(\mathbf{A}_t^{(n)})$, and executes only the first action. Candidate generation and scoring are described in Section III-D.

B. Dynamic Map-Conditioned Neighborhood Encoding

Dense racing traffic changes rapidly. A predetermined vehicle ordering may include irrelevant vehicles or miss the currently decisive opponent. DNQ-DLC therefore selects interaction-relevant vehicles online, rebuilds the local interaction graph, and tensorizes that graph through bounded masked relation slots.

Track-conditioned self geometry. For any vehicle $q \in \{0\} \cup \mathcal{I}_t$, let $c(\eta_t^q)$ be the closest centerline point to the vehicle, and let $\hat{\mathbf{t}}(\eta_t^q)$ and $\hat{\mathbf{n}}(\eta_t^q)$ be the corresponding local tangent and normal unit vectors. The track-relative self geometry is

$$\begin{aligned} \Delta x_{c,t}^q &= p_{x,t}^q - c_x(\eta_t^q), & \Delta y_{c,t}^q &= p_{y,t}^q - c_y(\eta_t^q), \\ \ell_t^q &= \frac{\langle p_t^q - c(\eta_t^q), \hat{\mathbf{n}}(\eta_t^q) \rangle}{w_t^q + \epsilon}, & e_{\psi,t}^q &= \text{wrap}(\psi_c(\eta_t^q) - \psi_t^q). \end{aligned} \quad (5)$$

This sign convention follows the submitted telemetry builder, where the heading-error features are represented by $\sin(\psi_c(\eta_t^q) - \psi_t^q)$ and $\cos(\psi_c(\eta_t^q) - \psi_t^q)$. The self feature block ϕ_t^q combines these track-conditioned quantities with normalized global position, velocity, progress, and driving-status flags. Its exact layout is reported in Supplementary Section I-D and Table SII.

For the controlled ego row, we also define the local map context

$$\mathcal{F}_t^0 = \{c(\eta_t^0), \hat{\mathbf{t}}(\eta_t^0), \hat{\mathbf{n}}(\eta_t^0), w_t^0, \kappa_t^0\}, \quad (7)$$

where w_t^0 is the local track-width scale and κ_t^0 is a look-ahead curvature feature.

Map-conditioned relation and interaction selection. For each surrounding vehicle $j \in \mathcal{I}_t$, DNQ-DLC forms the ego-neighbor physical relation vector

$$r_t^{0j} = [\Delta f_{b,t}^{0j}, \Delta l_{b,t}^{0j}, \Delta v_{x,t}^{0j}, \Delta v_{y,t}^{0j}, d_t^j, \sin(\Delta \psi_t^{0j}), \cos(\Delta \psi_t^{0j})]^\top. \quad (8)$$

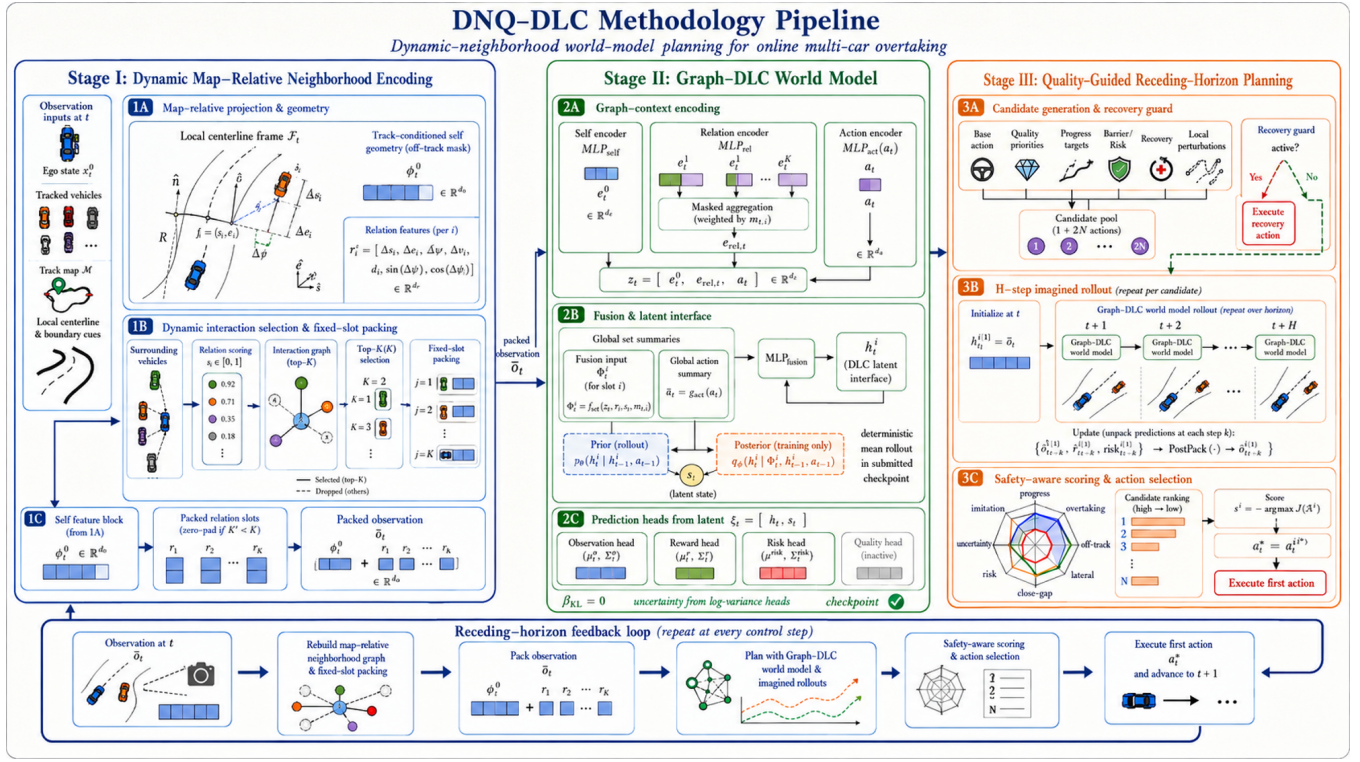


Fig. 1. Overall DNQ-DLC methodology pipeline. Stage I projects the ego and surrounding vehicles into a map-relative frame, ranks interaction-relevant neighbors, and packs the selected relations into a bounded masked tensor. Stage II encodes self, relation, and action features in the Graph-DLC world model and predicts latent state, reward, risk, and uncertainty. Stage III generates candidate actions, evaluates H -step imagined rollouts, applies safety-quality scoring and recovery logic, and executes the first action. The lower feedback loop shows that the neighborhood graph is rebuilt after every real observation.

Here $\Delta f_{b,t}^{0j}$ and $\Delta l_{b,t}^{0j}$ are the ego-body forward and left displacements used by the submitted telemetry builder, $\Delta v_{x,t}^{0j}$ and $\Delta v_{y,t}^{0j}$ are normalized global velocity differences, d_t^j is the normalized ego-neighbor distance, and $\Delta \psi_t^{0j} = \psi_t^j - \psi_t^0$ is the pairwise heading difference. Thus, the relation slot is map-conditioned through the ego state and local map context, but it is not a purely centerline-frame relation in every channel.

The interaction relevance score is written abstractly as

$$\rho_j(t) = \text{Score}_{\text{int}}(r_t^{0j}, \mathcal{F}_t^0), \quad (9)$$

and the eligible local neighborhood is

$$\mathcal{N}_t = \text{Gate}_{\text{int}}(\mathcal{I}_t, \{\rho_j(t)\}_j, \mathcal{F}_t^0). \quad (10)$$

All reported experiments use the implemented runtime interaction selector specified in Supplementary Section I-F (Runtime Interaction Selector). The abstract $\text{Score}_{\text{int}}$ and Gate_{int} notation is used to describe the general role of relevance-based interaction selection in the method pipeline.

Bounded graph tensorization and validity mask. The selected ego-centric graph is denoted by

$$G_t^0 = (V_t^0, \mathcal{E}_t^0, \mathbf{m}_t^0), \quad (11)$$

where V_t^0 contains the controlled ego vehicle and selected neighbors, $\mathcal{E}_t^0$ contains ego-neighbor relation edges, and $\mathbf{m}_t^0$ is the deterministic validity mask generated by

bounded slot tensorization. Let $K_{\max}$ denote the relation capacity used to batch the runtime graph. Vehicles in $\mathcal{N}_t$ are sorted by $\rho_j(t)$, and

$$j_t^k = \text{TopK}_k(\mathcal{N}_t; \rho), \quad k = 1, \dots, K_{\max}, \quad (12)$$

denotes the k -th highest-ranked valid neighbor when it exists. The validity mask is

$$m_t^{0,k} = \mathbb{I}[k \leq \min(|\mathcal{N}_t|, K_{\max})], \quad (13)$$

and the packed physical relation is

$$\tilde{r}_{\text{phy},t}^{0,k} = \begin{cases} r_t^{0j_t^k}, & m_t^{0,k} = 1, \\ \mathbf{0}, & m_t^{0,k} = 0. \end{cases} \quad (14)$$

Thus,

$$\mathbf{m}_t^0 = \{m_t^{0,k}\}_{k=1}^{K_{\max}}, \quad y_t^{0,k} = [\tilde{r}_{\text{phy},t}^{0,k}, m_t^{0,k}]. \quad (15)$$

The resulting bounded tensor representation of the runtime graph is

$$\bar{o}_t = \text{Pack}(G_t^0, \mathcal{M}) = [\phi_t^0, y_t^{0,1}, \dots, y_t^{0,K_{\max}}]. \quad (16)$$

Padded slots contain zero physical relation features and have zero validity mask. When $|\mathcal{N}_t| > K_{\max}$, only the $K_{\max}$ highest-ranked vehicles are represented by the network at that decision step. Thus, “dynamic neighborhood” means that the physical interaction graph and represented vehicle identities are rebuilt online; bounded masked tensorization is used for computation and is not

a predetermined vehicle topology. The submitted slot capacity and exact packed-observation layout are given in Supplementary Section I-B (Packed Observation Layout).

Graph-context encoding. For row i , the self feature block $\phi_t^i \in \mathbb{R}^{d_\phi}$ is encoded as

$$e_{\text{self},t}^i = \text{MLP}_{\text{ego}}(\phi_t^i). \quad (17)$$

The physical part of the k -th relation slot is encoded as

$$e_{\text{rel},t}^{i,k} = \text{MLP}_{\text{rel}}(\tilde{r}_{\text{phy},t}^{i,k}). \quad (18)$$

The validity mask is used for masked attention:

$$b_t^{i,k} = \text{Linear}_{\text{att}}(e_{\text{rel},t}^{i,k}), \quad (19)$$

$$\alpha_t^{i,k} = \frac{m_t^{i,k} \exp(b_t^{i,k})}{\epsilon + \sum_{\ell=1}^{K_{\max}} m_t^{i,\ell} \exp(b_t^{i,\ell})}. \quad (20)$$

The relation context is

$$\bar{e}_{\text{rel},t}^i = \sum_{k=1}^{K_{\max}} \alpha_t^{i,k} e_{\text{rel},t}^{i,k}. \quad (21)$$

The graph-context embedding for row i is

$$z_t^i = [e_{\text{self},t}^i, \bar{e}_{\text{rel},t}^i]. \quad (22)$$

For the controlled ego row, $z_t \equiv z_t^0$. Thus, $\bar{o}_t$ is the pre-encoding packed observation, whereas z_t is the encoded graph-context representation used by the world model.

C. Graph-DLC World Model

The Graph-DLC world model predicts short-horizon consequences under candidate actions. Its current observation input is $\bar{o}_t$, which is encoded into z_t , and its control input is a candidate action a_t . The action is encoded as

$$e_{\text{act},t} = \text{MLP}_{\text{act}}(a_t). \quad (23)$$

The implementation evaluates valid agent-centered rows in a shared batch. Let

$$\mathcal{R}_t = \{0\} \cup \mathcal{I}_t \quad (24)$$

be the row set, where row 0 corresponds to the controlled ego vehicle. For row i , the row-wise action embedding is

$$e_{\text{act},t}^i = \text{MLP}_{\text{act}}(a_t^i). \quad (25)$$

The global self and action summaries are

$$\bar{e}_{\text{self},t} = \frac{1}{|\mathcal{R}_t|} \sum_{i \in \mathcal{R}_t} e_{\text{self},t}^i, \quad \bar{e}_{\text{act},t} = \frac{1}{|\mathcal{R}_t|} \sum_{i \in \mathcal{R}_t} e_{\text{act},t}^i. \quad (26)$$

The fusion input and transition hidden state are

$$\Phi_t^i = [z_t^i, e_{\text{act},t}^i, \bar{e}_{\text{self},t}, \bar{e}_{\text{act},t}], \quad (27)$$

$$h_t^i = \text{MLP}_{\text{fusion}}(\Phi_t^i). \quad (28)$$

Since $z_t^i = [e_{\text{self},t}^i, \bar{e}_{\text{rel},t}^i]$, this is equivalent to the five-summary input $[e_{\text{self},t}^i, \bar{e}_{\text{rel},t}^i, e_{\text{act},t}^i, \bar{e}_{\text{self},t}, \bar{e}_{\text{act},t}]$. In the decision layer, $h_t \equiv h_t^0$. The treatment of non-controlled row actions is specified in Supplementary Section I-F (Runtime Interaction Selector).

DNQ-DLC keeps a DLC-style latent interface:

$$p_\theta(s_t|h_t) = \mathcal{N}(\mu_t^p, \text{diag}((\sigma_t^p)^2)), \quad (29)$$

$$q_\phi(s_t|h_t, \bar{o}_{t+1}) = \mathcal{N}(\mu_t^q, \text{diag}((\sigma_t^q)^2)). \quad (30)$$

The posterior uses $\bar{o}_{t+1}$ only during training. During online candidate ranking, the submitted checkpoint uses deterministic mean rollouts and the deterministic latent state

$$s_t^{\text{eval}} = \mu_t^p. \quad (31)$$

Therefore, the uncertainty penalty is derived from the observation and reward log-variance heads rather than from KL-regularized latent sampling.

The prediction heads operate on

$$\xi_t = [h_t, s_t]. \quad (32)$$

They are implemented as linear heads:

$$(\mu_{t+1|t}^o, \lambda_{t+1|t}^o) = \text{Linear}_{\text{obs}}(\xi_t), \quad (33)$$

$$(\mu_{t+1|t}^r, \lambda_{t+1|t}^r) = \text{Linear}_{\text{rew}}(\xi_t), \quad (34)$$

$$z_{t+1|t}^x = \text{Linear}_{\text{risk}}(\xi_t), \quad \hat{\chi}_{t+1|t} = \sigma(z_{t+1|t}^x). \quad (35)$$

Here $\mu_{t+1|t}^o$ and $\lambda_{t+1|t}^o$ parameterize the next packed-observation distribution, $\mu_{t+1|t}^r$ and $\lambda_{t+1|t}^r$ parameterize the reward distribution, and $\hat{\chi}_{t+1|t}$ is a boundary-risk probability. The implementation also contains an auxiliary quality head. Its loss weight is zero in the submitted checkpoint, so the head does not contribute to quality guidance (Supplementary Section II-C).

The general world-model objective is

$$\begin{aligned} \mathcal{L}_{\text{wm}} = \mathbb{E}_{q_\phi} [\mathcal{L}_o + \mathcal{L}_r + \lambda_\chi \text{BCE}(\chi_{t+1}, \hat{\chi}_{t+1|t}) + \lambda_q \mathcal{L}_q] \\ + \beta_{\text{KL}} D_{\text{KL}}(q_\phi \| p_\theta). \end{aligned} \quad (36)$$

In the submitted checkpoint, $\lambda_q = 0$ and $\beta_{\text{KL}} = 0$. Therefore, the auxiliary quality loss and KL regularization are inactive in world-model training. Quality guidance in the planner comes from the separately trained quality-biased actor, overtaking and recovery templates, and online safety-efficiency scoring terms. The Gaussian likelihoods, dataset size and validation splits are reported in Supplementary Section II-A. The loss definitions and checkpoint-specific interpretation are given in Supplementary Section II-C, and actor training is detailed in Supplementary Section II-D.

D. Quality-Guided Receding-Horizon Planning

DNQ-DLC converts prediction into control by combining learned actor proposals, analytic overtaking templates, recovery actions, safety actions, and local perturbations. The analytic templates and recovery guard are not claimed as learned modules; they serve as structured racing priors that expand the candidate set and prevent degenerate learned proposals under boundary or recovery conditions. Before candidate scoring, a deterministic recovery guard is evaluated. If the guard is active, DNQ-DLC executes the recovery action directly; otherwise it ranks candidate

rollouts by the score below. Candidate-pool construction, duplicate removal, passing templates, braking templates, and the recovery guard are reported in Supplementary Section III (Candidate Generation and Scoring).

The learned proposals are

$$a_t^{\text{base}} = \text{ActorMLP}_{\text{base}}(\bar{o}_t), \quad a_t^{\text{qual}} = \text{ActorMLP}_{\text{qual}}(\bar{o}_t). \quad (37)$$

Thus, $\bar{o}_t$ serves three roles: it is encoded into graph context z_t , consumed by the proposal actors, and recursively used as the state interface during imagined rollout.

Analytic passing proposals are activated when the local geometry admits an attack:

$$I_{\text{pass}} = \mathbb{I}[|\ell_t^0| < \ell_{\text{att}}, \cos(e_{\psi,t}^0) > \eta_{\text{att}}, n_{\text{front}} + n_{\text{side}} > 0]. \quad (38)$$

When activated, left- and right-side passing actions are generated as

$$a_t^{\text{pass}}(s, \Delta) = [\text{clip}(s\Delta - \lambda_{\psi}e_{\psi,t}^0, \delta_{\min}, \delta_{\max}), u_{\text{pass}}(\kappa_t^0), 0]^\top. \quad (39)$$

The concrete trigger values and template constants are provided in Supplementary Section III (Candidate Generation and Scoring).

For candidate n , the imagined rollout starts from

$$\hat{o}_{t|t}^{(n)} = \bar{o}_t, \quad \hat{a}_{t|t}^{(n)} = a_t^{(n)}. \quad (40)$$

For future imagined steps $k > 0$, the continuation action is

$$\begin{aligned} \hat{o}_{t+k|t}^{(n)} = & (1 - \alpha_q) \text{ActorMLP}_{\text{base}}(\hat{o}_{t+k|t}^{(n)}) \\ & + \alpha_q \text{ActorMLP}_{\text{qual}}(\hat{o}_{t+k|t}^{(n)}). \end{aligned} \quad (41)$$

The same Graph-DLC pipeline is applied to $\hat{o}_{t+k|t}^{(n)}$ and $\hat{a}_{t+k|t}^{(n)}$, and the next imagined packed observation is updated by

$$\hat{o}_{t+k+1|t}^{(n)} = \text{PostPack}(\mu_{t+k+1|t}^{o,(n)}, \hat{\mathbf{m}}_{t+k|t}^{0,(n)}). \quad (42)$$

In the submitted short-horizon rollout, relation-slot identities are not explicitly tracked as persistent physical agents. The predicted mean provides continuous self and relation features, while the validity mask channels are post-processed by $\text{PostPack}(\cdot)$ for the next masked aggregation. At the next real simulator step, the neighborhood is rebuilt from telemetry and repacked by $\text{Pack}(\cdot)$. Implementation details for $\text{PostPack}(\cdot)$ are provided in Supplementary Section III-A (Mask Handling During Imagined Rollout).

Each candidate rollout is scored by

$$\begin{aligned} J(\mathbf{A}_t^{(n)}) = & \sum_{k=0}^{H-1} \gamma^k [\lambda_{\text{prog}} R_{\text{prog}}^{(n,k)} + \lambda_{\text{ovt}} R_{\text{ovt}}^{(n,k)} \\ & - \lambda_{\text{off}} C_{\text{off}}^{(n,k)} - \lambda_{\text{lat}} C_{\text{lat}}^{(n,k)} \\ & - \lambda_{\text{close}} C_{\text{close}}^{(n,k)} - \lambda_{\text{risk}} C_{\text{risk}}^{(n,k)} \\ & - \lambda_{\text{unc}} C_{\text{unc}}^{(n,k)} - \lambda_{\text{imit}} C_{\text{imit}}^{(n,k)}]. \end{aligned} \quad (43)$$

Full scoring definitions are given in Supplementary Section III-B (Scoring Terms).

The selected action is

$$n^* = \arg \max_n J(\mathbf{A}_t^{(n)}), \quad a_t^* = a_t^{(n^*)}. \quad (44)$$

DNQ-DLC applies a_t^* , observes telemetry at $t+1$, rebuilds G_{t+1}^0 and $\bar{o}_{t+1}$, and replans. Since this layer uses weighted penalties rather than formal hard constraints, we describe it as safety-aware multi-objective scoring rather than strict safety-constrained optimization.

IV. Experiments and Results

A. Experimental Setup and Metrics

The primary benchmark contains 200 matched cases per algorithm over $N \in \{4, 5, 6, 7, 8\}$ vehicles. The $N = 4$ –6 cases match the training vehicle-count range, whereas $N = 7$ –8 tests vehicle-count extrapolation. Frozen-checkpoint validation evaluates the world model and quality actor on episode-disjoint procedural, Hairpin, and S-Curve data. The component study uses matched randomized cases in which start slots and vehicle spacing change. Every policy observes all vehicles and selects $K = 3$ neighbors; failed cases are retained.

The benchmark compares DNQ-DLC with Rule Expert, Safety Rule, PPO, SAC, TD3 [31]–[33], and three DLC variants. Rule Expert combines lane following, adaptive driving, and passing modes. Safety Rule uses more conservative barrier and traffic-density thresholds. Their implementation is described in Supplementary Section II-B.

DLC-IT, DLC-JT, and DLC-JTO denote the individual-transition, joint-transition, and joint-transition-plus-observer variants of Schwarting et al. [15]. We retrained each variant from four seeds using 120 episodes, 96000 transitions, and 80 updates. A frozen checkpoint was chosen by a predefined round-robin validation rule. These baselines use fixed but non-identical training budgets and architectures. Table II tests all four final DLC-JTO checkpoints on eight new matched cases.

The legacy PPO, SAC, and TD3 checkpoints use one seed and 50k environment steps. The training-budget analysis adds three 200k-step seeds for each RL family and evaluates them on 12 new randomized $N = 4$ cases. Full training budgets and checkpoint rules are given in Supplementary Section II-B.

The analysis designates binary overtaking success as the primary closed-loop endpoint. On-track and desirable overtaking, rank gain, target grass exposure, the contact proxy, and conditional completion time are secondary endpoints. Episode-level on-track and desirable proportions use all cases, with zero assigned when a case has no completed overtake. Completion time and target-grass summaries are conditioned on successful cases and always report their sample counts.

Because every algorithm was evaluated on the same 200 case identifiers, primary contrasts use the paired success-rate difference. We report a 50,000-resample paired case-bootstrap 95% confidence interval and a two-sided exact McNemar test. Holm adjustment covers the six reported

baseline contrasts. Absolute success rates retain Wilson intervals for scale, and vehicle-count strata are reported as sensitivity analyses rather than independent replications. Model validation reports state RMSE, reward MAE, risk AUROC, risk AUPRC, Brier score, expected calibration error, and H -step state RMSE. Metric definitions and statistical summaries are given in Supplementary Section IV-A. The matched primary and randomized attribution protocols are described in Supplementary Sections IV-B and IV-C, respectively.

B. Experiment 1: Basic Effectiveness

Figure 2 reports the 200 matched cases at the aggregate and vehicle-count levels. Table I gives the paired primary-endpoint tests for the same cases.

Across the same 200 matched cases, DNQ-DLC achieved 85.5% success (171/200), compared with 71.5% for Rule Expert (143/200). The paired increase was 0.140 (95% paired bootstrap CI, 0.065–0.215; exact McNemar $p = 0.000617$). Both controllers succeeded in 125 cases, only DNQ-DLC succeeded in 46, only Rule Expert succeeded in 18, and neither succeeded in 11. Relative to Safety Rule, the paired increase was 0.155 (95% CI, 0.085–0.225; Holm-adjusted $p = 0.000066$). Relative to DLC-JTO, the paired increase was 0.765 (95% CI, 0.700–0.830; Holm-adjusted $p < 10^{-40}$). Table I reports the complete matched comparison family.

Secondary outcomes show a clear trade-off. Conditional mean completion time was 97.8 steps for DNQ-DLC, 126.0 for Rule Expert, and 131.1 for Safety Rule. These estimates use different successful-case sets and are not an all-case censored-time comparison. Mean desirable overtaking was 42.0% for DNQ-DLC and 38.1% for Rule Expert, but mean target grass exposure among successful cases was 62.9% for DNQ-DLC and 39.0% for Rule Expert. Overtaking success is therefore an effectiveness endpoint, not a surrogate for global safety or track adherence.

The effect differed by vehicle count. Within $N = 4$ –6, success was 86.7% for DNQ-DLC and 81.7% for Rule Expert. The paired difference was 0.050 (95% CI, -0.033 – 0.133 ; exact McNemar $p = 0.327$), which does not establish a difference in this stratum. For $N = 7$ –8, success was 83.8% for DNQ-DLC and 56.3% for Rule Expert. The paired difference was 0.275 (95% CI, 0.138–0.413; exact McNemar $p = 0.000472$). This result supports vehicle-count extrapolation within the tested simulator range, not unrestricted out-of-distribution generalization.

Table II reports the checkpoint-sensitivity analysis. The four final DLC-JTO checkpoints were evaluated on eight new matched $N = 4$ cases. No checkpoint was selected on these cases.

DNQ-DLC succeeded in 7/8 cases, Rule Expert in 6/8, and every DLC-JTO checkpoint in 0/8. The paired DNQ-DLC minus DLC-JTO difference was 0.875 for each checkpoint (95% CI, 0.625–1.000). The individual exact McNemar value was 0.0156, while Holm adjustment across the four checkpoint comparisons gave 0.0625. The small

evaluation set makes this a sensitivity result rather than a second confirmatory benchmark.

C. Training-Budget Sensitivity: Multi-seed RL Baselines

Table III summarizes 12 evaluation cases for each of three independently trained PPO, SAC, and TD3 policies, together with the fixed DNQ-DLC and rule-based references.

Longer RL training improves several baselines, but no controller leads on every metric in Table III. The unchanged DNQ-DLC controller completes an overtake in all 12 cases and has substantially lower grass exposure than PPO, SAC, and Rule Expert. However, Rule Expert and PPO have much higher desirable-overtaking rates, Safety Rule has higher mean rank gain, and TD3 has lower mean grass exposure. The RL family ranges also reveal material training-seed variability. These results reinforce the central trade-off: DNQ-DLC prioritizes completion in this short horizon, but does not dominate behavior-quality endpoints.

D. Experiment 2: Frozen World-Model Validation

Table IV reports one-step and ten-step prediction results for three episode-disjoint validation strata, with eight episodes and 4800 transitions in each stratum.

Table IV shows that the frozen transition model is most accurate in the procedural $N = 4$ –6 stratum. One-step state RMSE approximately doubles on Hairpin with eight vehicles, and $H = 10$ error increases in both out-of-distribution strata. The risk head remains highly discriminative and well calibrated for its implemented boundary-risk target (ECE 0.0006, 0.0080, and 0.0036, respectively), but this does not establish collision prediction or real-world safety. The risk head remains useful for short-horizon ranking, but state prediction transfers less well over longer horizons.

E. Experiment 3: Quality-Proposal Validation

Table V compares the quality actor with the training-set mean-action predictor on newly collected accepted-event states from the three validation strata.

The quality actor outperforms the mean-action predictor only in the procedural training-range stratum. Its error is much higher on both unseen $N = 8$ tracks, mainly because it predicts too much throttle. The actor is therefore useful as an additional proposal, but it does not show out-of-distribution generalization. Closed-loop results apply to the complete controller rather than to this actor alone.

F. Experiment 4: Same-Track Interaction Sequence

Figure 3 shows synchronized top-down and ego views from one shared-scene run. DNQ-DLC starts behind Rule Expert, TD3, and DLC-JTO in the same simulator instance.

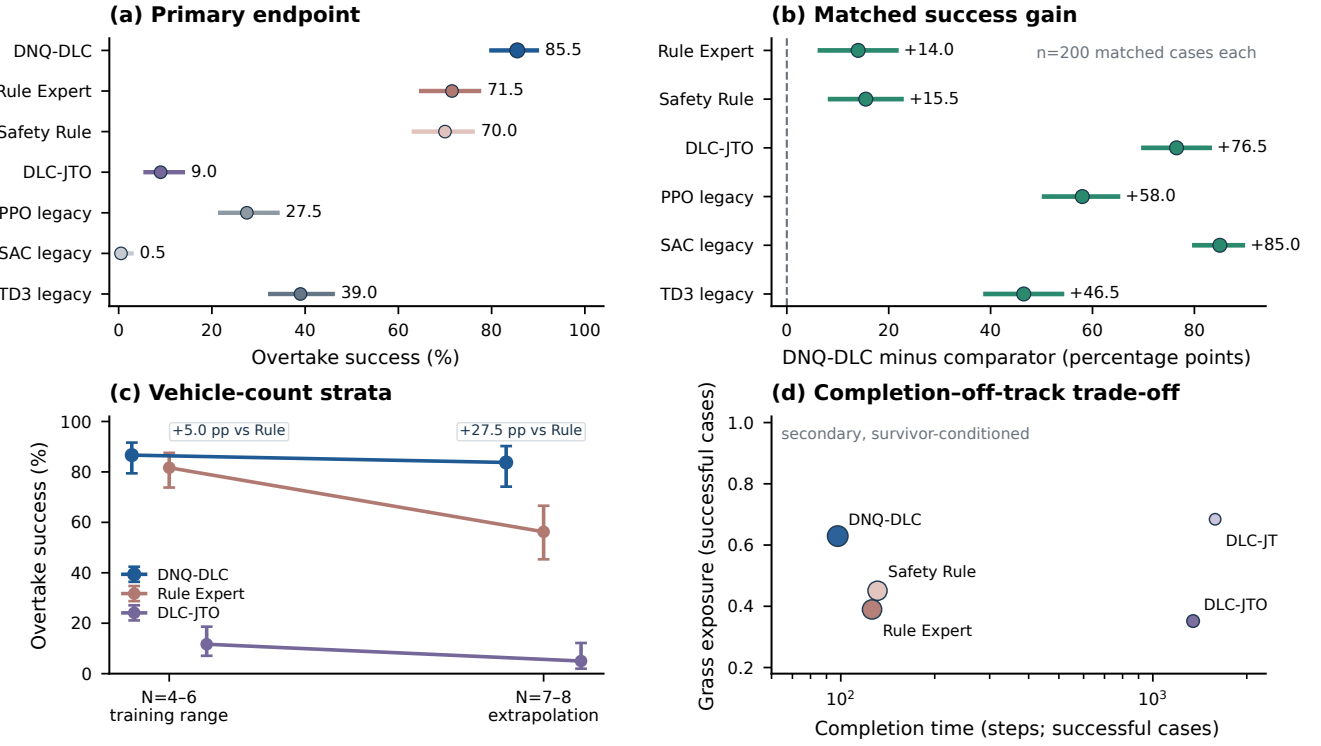


Fig. 2. Primary overtaking evidence on the 200-case matched benchmark. (a) Absolute success rates with Wilson 95% confidence intervals. (b) Paired DNQ-DLC minus comparator differences with 50,000-resample paired bootstrap intervals. (c) Training-range and vehicle-count-extrapolation strata; labels report the paired difference from Rule Expert, with the corresponding intervals reported in Table I. (d) Completion time and grass exposure among successful cases. Panel (d) is survivor-conditioned and descriptive rather than an all-case safety comparison; marker area indicates the successful-case count.

TABLE I
Matched primary-endpoint comparisons on the 200-case benchmark

Method	Success	DNQ-DLC minus method [95% CI]	Discordant	p_{Holm}
DNQ-DLC	171/200	Reference	—	—
Rule Expert	143/200	0.140 [0.065, 0.215]	46/18	0.000617
Safety Rule	140/200	0.155 [0.085, 0.225]	43/12	0.000066
DLC-JTO	18/200	0.765 [0.700, 0.830]	157/4	$< 10^{-40}$
PPO legacy	55/200	0.580 [0.505, 0.650]	119/3	$< 10^{-30}$
SAC legacy	1/200	0.850 [0.800, 0.895]	170/0	$< 10^{-50}$
TD3 legacy	78/200	0.465 [0.390, 0.540]	98/5	$< 10^{-22}$

Note: Effects are DNQ-DLC minus comparator. Confidence intervals use 50,000 paired case-bootstrap resamples. Discordant pairs are DNQ-DLC-only/comparator-only successes. McNemar p values are Holm-adjusted. Legacy RL checkpoints are not training-budget matched. Boldface marks the highest success count.

The sequence makes the maneuver order visible without serving as a baseline estimate. The four controllers do not have matched training budgets, and the case was chosen for qualitative inspection. The full video is available at <https://chocho2222.github.io/DNQ-DLC/#videos>.

G. Experiment 5: Mechanism Explanation

Figure 4 combines one annotated overtaking sequence with the 32-case graph-repacking summary and paired component effects.

Panels (a)–(c) show A9 moving from one tile behind A4 to equal progress and then two tiles ahead. The enlarged views make the change in relative progress clear. Panel (d) shows that all 32 full-controller cases changed their

ranked top-three relations, with a median of 43 switches per case. The $N = 8$ cases also used more distinct relation sets than the $N = 6$ cases. Panel (e) supports effects from risk/uncertainty scoring, safety-quality scoring, and the quality proposal's reduction in grass exposure. The dynamic-selector intervals cross zero in this matrix.

TABLE II
DLC-JTO checkpoint sensitivity on eight new matched $N = 4$ cases

Controller/checkpoint	Success [95% Wilson CI]	Desirable	Target grass	Rank gain
DNQ-DLC fixed	7/8 [0.529, 0.978]	0.175	0.536	0.750
Rule Expert fixed	6/8 [0.409, 0.929]	0.438	0.566	1.500
DLC-JTO train seed 0	0/8 [0.000, 0.324]	0.000	0.529	0.000
DLC-JTO train seed 1	0/8 [0.000, 0.324]	0.000	0.528	0.000
DLC-JTO train seed 2	0/8 [0.000, 0.324]	0.000	0.791	0.000
DLC-JTO train seed 3	0/8 [0.000, 0.324]	0.000	0.395	0.000

Note: DNQ-DLC and Rule Expert are fixed references. Boldface marks the best point estimate in each directly comparable performance column; it does not indicate statistical significance.

TABLE III
Three-seed evaluation of 200k-step RL baselines

Family	Success	Desirable	Grass	Rank gain
DNQ-DLC	1.000	0.054	0.115	2.167
Rule Expert	0.917	0.417	0.220	2.000
Safety Rule	0.917	0.347	0.151	2.333
PPO	0.694 [0.667, 0.750]	0.410 [0.333, 0.500]	0.876 [0.743, 0.957]	0.278 [0.000, 0.750]
SAC	0.889 [0.833, 0.917]	0.287 [0.000, 0.444]	0.236 [0.161, 0.379]	1.250 [0.500, 2.250]
TD3	0.806 [0.750, 0.917]	0.148 [0.028, 0.333]	0.080 [0.000, 0.192]	0.917 [0.083, 2.500]

Note: Evaluation uses 12 new procedural $N = 4$ cases. RL entries are means and ranges across three training seeds; ranges are not confidence intervals. DNQ-DLC and rule controllers are fixed. Boldface marks the best mean point estimate.

TABLE IV
Episode-disjoint validation of the frozen graph world model

Stratum	One-step state RMSE	Reward MAE	Risk prevalence	AUROC	AUPRC	$H = 10$ RMSE
Procedural, $N = 4-6$	0.124	0.198	0.005	0.999	0.982	0.239
Hairpin, $N = 8$	0.248	0.569	0.353	0.996	0.995	0.409
S-Curve, $N = 8$	0.176	0.535	0.339	0.999	0.999	0.370

Note: The risk target is the implemented boundary/unstable-driving label, not physical collision probability. Each stratum contains eight episodes and 4800 transitions. Boldface marks the lowest prediction error or highest discrimination value; risk prevalence is not ranked.

TABLE V
Episode-disjoint validation of the quality-proposal actor

Stratum	Episodes with samples	Samples	Accepted events	Actor MSE	Mean-action MSE
Procedural, $N = 4-6$	3/8	250	3	0.015	0.031
Hairpin, $N = 8$	8/8	1446	16	0.139	0.028
S-Curve, $N = 8$	8/8	758	8	0.114	0.029

Note: Validation uses newly collected accepted-event states. Actor MSE is compared with a constant predictor equal to the mean action in the training set. Boldface marks the lower MSE within each stratum.

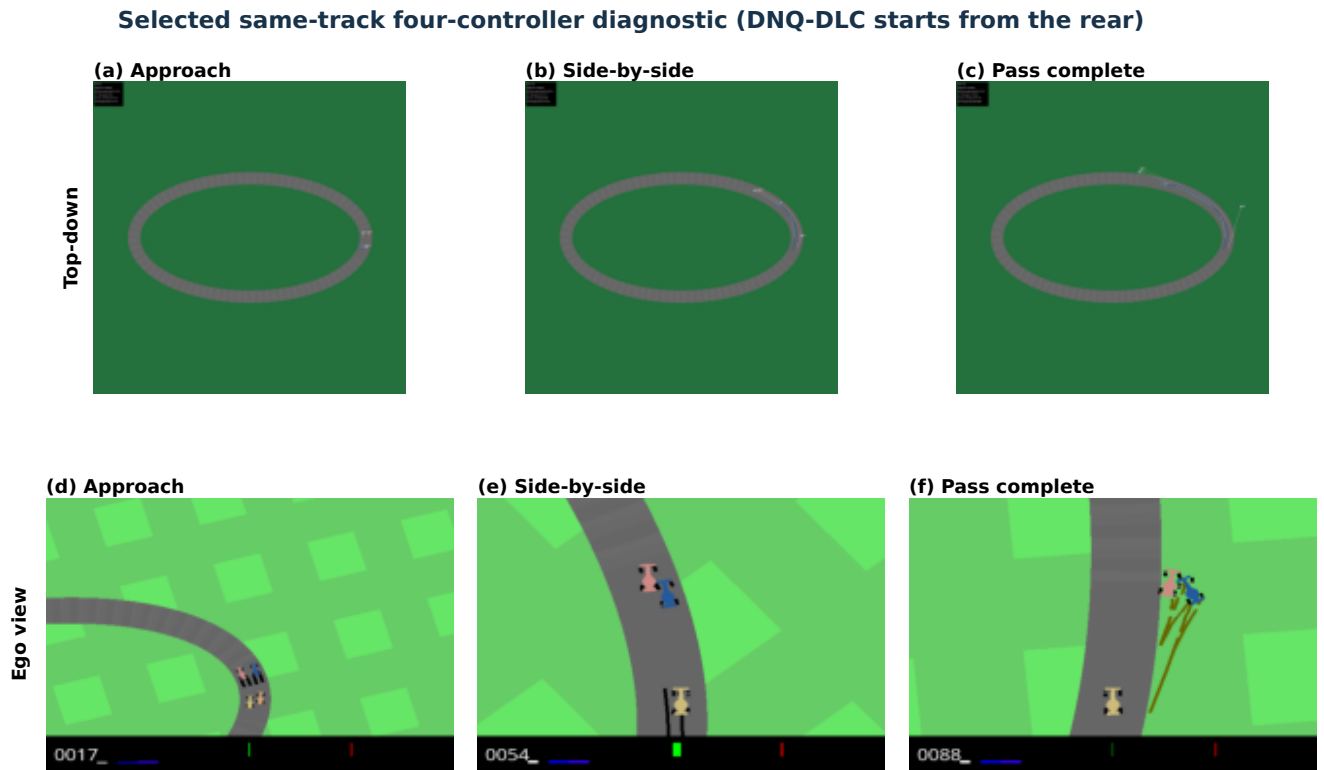


Fig. 3. Same-track qualitative interaction sequence. The top and bottom rows show synchronized top-down and ego views during approach, side-by-side interaction, and pass completion. DNQ-DLC starts from the rear and shares the scene with Rule Expert, TD3, and DLC-JTO. The sequence illustrates one maneuver and is not used for population-level comparison.

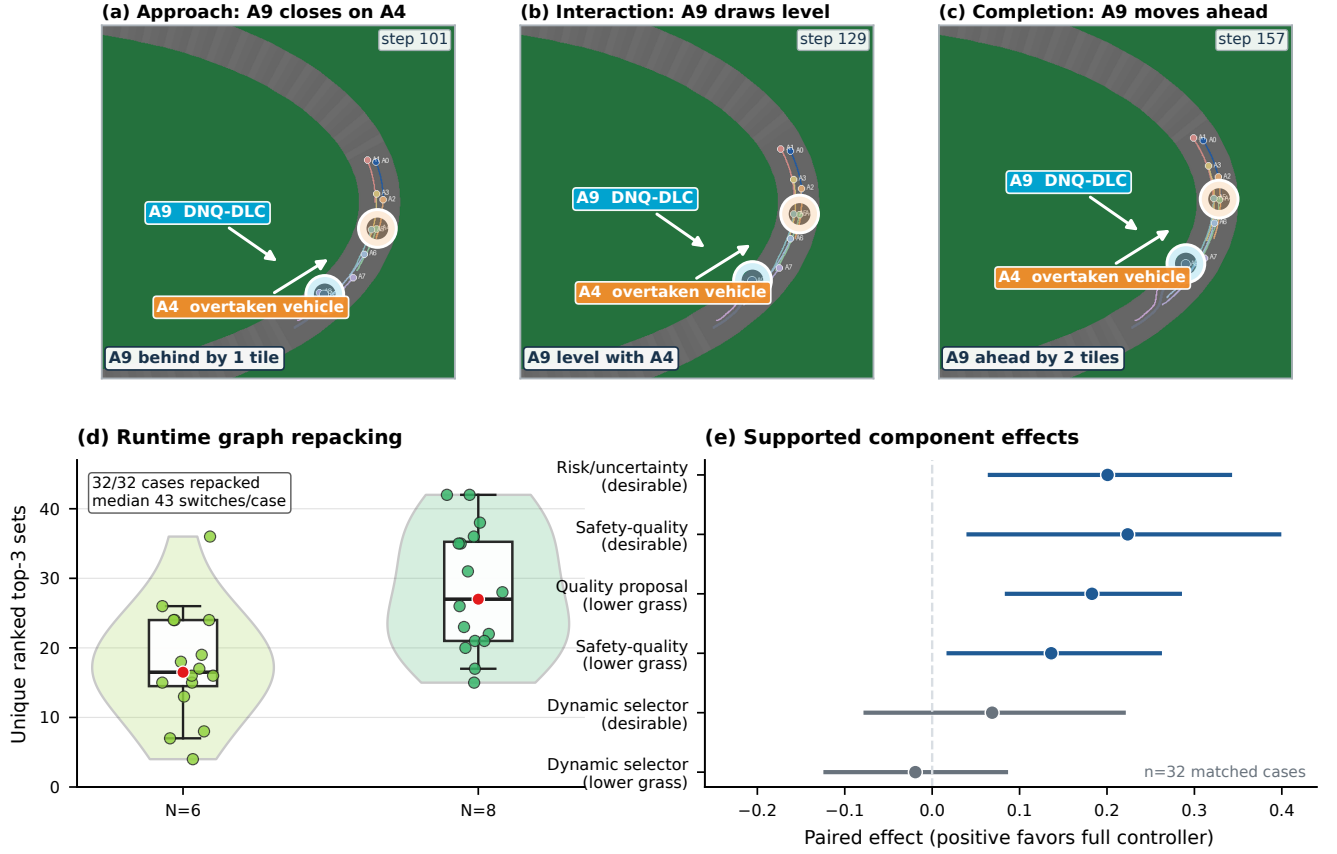


Fig. 4. Runtime graph repacking and matched component evidence. (a)–(c) Magnified top-down views of one ten-vehicle overtaking event. A9 closes on A4 at step 101, reaches equal lap-tile progress at step 129, and leads by two tiles at step 157. Cyan and orange annotations identify A9 and A4, respectively. (d) Unique ranked top-three sets across the 32 full-controller cases in the randomized $N = 6/8$ matrix. Violin envelopes show density, boxes show the interquartile range and median, and points show individual cases. (e) Paired full-controller effects for the same cases. Positive values favour the full controller; blue intervals exclude zero and grey intervals cross zero.

H. Experiment 6: Randomized Component Attribution

Table VI summarizes all 288 runs in the randomized $N = 6/8$ attribution matrix. Figure 4(e) shows the paired confidence intervals for the effects discussed below.

Table VI reports all prespecified randomized cases. Removing the learned risk and uncertainty costs reduces the desirable rate by a paired mean of 0.201 (95% bootstrap CI [0.066, 0.341]), although it produces one additional success. Removing all safety-quality terms reduces desirable behavior by 0.224 [0.042, 0.397] and raises grass exposure by 0.136 [0.019, 0.260]. The quality proposal has a measurable grass effect: removing it raises grass exposure by 0.183 [0.086, 0.283], while its desirable-rate and rank-gain effects remain uncertain at this sample size.

The selector variants all succeed in 30/32 cases, and their paired intervals overlap for desirable behavior, grass

exposure, and rank gain. Hard recovery also has little aggregate effect. Removing handcrafted candidates improves all four point estimates at $N = 6/8$. The higher-density test below gives a different result.

A separate $N = 10/12$ mixed-traffic test contains 32 matched cases. The full, fixed-identity, and nearest-neighbor controllers each succeed in 32/32 cases. Fixed identity has a higher desirable-rate point estimate (0.822 versus 0.703). The full controller has lower grass exposure than nearest neighbor (0.287 versus 0.396; paired difference -0.108 , 95% CI $[-0.202, -0.014]$). Removing handcrafted candidates reduces success to 27/32. Selector and candidate effects therefore depend on traffic density. Dynamic packing is supported as a changing interaction representation, not as a universally better selection policy.

V. Discussion

The main result is that the complete DNQ-DLC controller improves overtaking success relative to Rule Expert, Safety Rule, and the evaluated DLC and legacy RL baselines. This result applies to the full controller. It does not show that the dynamic selector alone caused the increase. Overtaking success measures pass completion, not overall safety. Rule Expert still has lower grass exposure and a higher desirable-overtaking rate within $N = 4-6$.

The supporting experiments explain this result more clearly. The frozen world model ranks short-horizon boundary risk well, but its prediction error increases on unseen $N = 8$ tracks. The quality actor works in the procedural training range but does not transfer to the two unseen $N = 8$ tracks. Randomized ablations support several quality-related terms, whereas the dynamic-selector intervals cross zero. Longer RL training also improves several baselines, and the methods occupy different parts of the completion-desirable-grass trade-off. DNQ-DLC has the highest completion in the 12-case $N = 4$ study, but it does not simultaneously provide the best desirable rate, grass rate, and rank gain.

The framework also clarifies the role of structured priors. Rule Expert is both a strong comparator and an internal dependency of the deployed hybrid planner through the anchor action. Handcrafted candidate templates, speed limits, and recovery triggers provide additional inductive bias. The system should therefore be interpreted as learned short-horizon ranking around structured racing priors, rather than as a fully learned replacement for rule-based control.

The evidence remains confined to simulation and does not establish real-vehicle safety, physical collision avoidance, tire-limit validity, or robustness to actuator delay. The implemented risk target is derived from off-track, backward-motion, and lateral-error telemetry rather than physical collision labels. The controller also assumes map-relative telemetry, leaving perception error, occlusion, localization drift, and asynchronous sensing outside the evaluated setting. Within imagined rollouts, relation slots remain fixed; the physical graph is rebuilt only after the next real simulator observation. The legacy E2/E3 sweeps

TABLE VI
Randomized component attribution across tracks and vehicle counts

Variant	Success (95% Wilson CI)	Desirable	Grass	Rank gain
Full controller	0.938 [0.799, 0.983]	0.799	0.465	2.562
Nearest-neighbor slots	0.938 [0.799, 0.983]	0.760	0.406	2.375
Fixed-identity slots	0.938 [0.799, 0.983]	0.730	0.446	2.500
w/o quality proposal	0.938 [0.799, 0.983]	0.669	0.648	1.906
w/o risk/uncertainty	0.969 [0.843, 0.994]	0.598	0.430	2.875
w/o all safety-quality terms	0.906 [0.758, 0.968]	0.575	0.602	2.344
w/o rule anchor	0.938 [0.799, 0.983]	0.701	0.377	2.812
w/o handcrafted candidates	0.969 [0.843, 0.994]	0.890	0.367	3.031
w/o hard recovery	0.938 [0.799, 0.983]	0.758	0.465	2.625

Note: The matched design uses Hairpin and S-Curve, $N = 6/8$, and eight seeds per stratum. All 288 runs completed. Grass is lower-is-better; the other metrics are higher-is-better. Boldface marks the best point estimate, not statistical significance.

are therefore retained as deterministic functional checks rather than independent scenario evidence.

Training-budget and opponent diversity also constrain comparison. The additional RL study covers three seeds, 200k steps, $N = 4$, and 12 final checkpoint evaluations without dense learning curves. The DLC suite uses four training seeds but a lower optimization budget than DNQ-DLC. The quality actor is trained from 24 filtered episodes and does not beat the mean-action predictor on two unseen $N = 8$ tracks. Moreover, the opponent set excludes adversarial blocking and human racecraft. Supplementary Section VI discusses these evidence limits.

VI. Conclusion and Future Work

This paper introduced DNQ-DLC, a hybrid world-model planner for online multi-car overtaking. At each real control step, the method reconstructs a map-conditioned interaction graph, selects the currently relevant neighboring vehicles, and packs them into a bounded masked graph tensor. A Graph-DLC model then predicts short-horizon outcomes, while a quality-guided receding-horizon planner ranks learned and structured action candidates. This design separates the changing physical interaction set from the fixed-dimensional neural interface.

On the 200-case matched benchmark, the complete controller achieved 85.5% overtaking success, compared with 71.5% for Rule Expert. The paired improvement was 14.0 percentage points (95% CI, 6.5–21.5 percentage points). The advantage was also observed in the tested $N = 7$ –8 vehicle-count extrapolation stratum. Frozen-model validation supports short-horizon boundary-risk ranking, and randomized attribution supports contributions from risk and safety-quality scoring. However, the method did not dominate every secondary endpoint, and the dynamic selector’s independent performance advantage was not established.

Future work will move the framework from telemetry-based simulation to closed-course real-vehicle overtaking experiments. The controller will be integrated with on-board perception, multi-vehicle tracking, localization, and a supervisory safety layer, and its latency and robustness will be evaluated under sensor noise and actuator delay. Hardware-in-the-loop tests will precede progressively more demanding multi-vehicle track experiments.

Data and Code Availability

The implementation, configurations, self-trained checkpoints, source data, figures, and simulation videos will be released after submission at <https://github.com/chocho2222/DNQ-DLC>. A frozen versioned release and archival DOI will be added after the public package is finalized.

References

- [1] P. R. Wurman, S. Barrett, K. Kawamoto, J. MacGlashan, K. Subramanian, T. J. Walsh, R. Capobianco, A. Devlic, F. Eckert, F. Fuchs, L. H. Gilpin, P. Khandelwal, V. Kompella, H. C. Lin, P. MacAlpine, D. Oller, T. Seno, C. Sherstan, M. D. Thomure, H. Aghabozorgi, L. Barrett, R. Douglas, N. Whitehead, A. Dutech, J. Smith, S. Okabe, M. Sridharan, and P. Stone, “Outracing champion Gran Turismo drivers with deep reinforcement learning,” *Nature*, vol. 602, pp. 223–228, 2022.
- [2] S. Feng, X. Yan, H. Sun, Y. Feng, and H. X. Liu, “Intelligent driving intelligence test for autonomous vehicles with naturalistic and adversarial environment,” *Nature Communications*, vol. 12, no. 1, p. 748, 2021.
- [3] H. X. Liu and S. Feng, “Curse of rarity for autonomous vehicles,” *Nature Communications*, vol. 15, no. 1, p. 4808, 2024.
- [4] J. Betz, T. Betz, F. Fent, M. Geisslinger, A. Heilmeier, L. Hermansdorfer, T. Herrmann, S. Huch, P. Karle, M. Lienkamp, B. Lohmann, F. Nobis, L. Ögretmen, M. Rowold, F. Sauerbeck, T. Stahl, R. Trauth, F. Werner, and A. Wischnewski, “Tum autonomous motorsport: An autonomous racing software for the indy autonomous challenge,” *Journal of Field Robotics*, 2023.
- [5] A. Wischnewski, M. Geisslinger, J. Betz, T. Betz, F. Fent, A. Heilmeier, L. Hermansdorfer, T. Herrmann, S. Huch, P. Karle, F. Nobis, L. Ögretmen, M. Rowold, F. Sauerbeck, T. Stahl, R. Trauth, M. Lienkamp, and B. Lohmann, “Indy autonomous challenge: Autonomous race cars at the handling limits,” *arXiv preprint arXiv:2202.03807*, 2022.
- [6] C. Jung, A. Finazzi, H. Seong, D. Lee, and S. Lee, “An autonomous racing system: Design, implementation, and analysis; team kaist at the iac,” *Field Robotics*, 2023.
- [7] A. Raji, A. Liniger, A. Giove, A. Toschi, N. Musiu, D. Morra, M. Verucchi, D. Caporale, and M. Bertogna, “Motion planning and control for multi-vehicle autonomous racing at high speeds,” *arXiv preprint arXiv:2207.11136*, 2022.
- [8] A. Wischnewski, T. Herrmann, F. Werner, and B. Lohmann, “A tube-mpc approach to autonomous multi-vehicle racing on high-speed ovals,” *IEEE Transactions on Intelligent Vehicles*, 2023.
- [9] J. Betz, A. Wischnewski, A. Heilmeier, F. Nobis, T. Stahl, L. Hermansdorfer, B. Lohmann, and M. Lienkamp, “Autonomous vehicles on the edge: A survey on autonomous vehicle racing,” *IEEE Open Journal of Intelligent Transportation Systems*, vol. 3, pp. 458–488, 2022.

- [10] B. R. Kiran, I. Sobh, V. Talpaert, P. Mannion, A. A. Sallab, S. Yogamani, and P. Pérez, "Deep reinforcement learning for autonomous driving: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 6, pp. 4909–4926, 2021.
- [11] E. Yurtsever, J. Lambert, A. Carballo, and K. Takeda, "A survey of autonomous driving: Common practices and emerging technologies," *IEEE Access*, vol. 8, pp. 58 443–58 469, 2020.
- [12] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering diverse domains through world models," *arXiv preprint arXiv:2301.04104*, 2023.
- [13] A. Hu, G. Corrado, N. Griffiths, Z. Murez, C. Gurau, H. Yeo, A. Kendall, R. Cipolla, and J. Shotton, "Mile: Model-based imitation learning for urban driving," 2022, *arXiv preprint arXiv:2210.07729*.
- [14] A. Hu, L. Russell, H. Yeo, Z. Murez, G. Fedoseev, A. Kendall, J. Shotton, and G. Corrado, "Gaia-1: A generative world model for autonomous driving," 2023, *arXiv preprint arXiv:2309.17080*.
- [15] W. Schwarting, A. Pierson, J. Alonso-Mora, S. Karaman, and D. Rus, "Deep latent competition: Learning to race using visual control policies in latent space," 2021, *arXiv preprint arXiv:2102.09812*.
- [16] S. Hoffmann, S. Sagmeister, T. Betz, J. Bongard, S. Büttner, D. Ebner, D. Esser, G. Jank, S. Goblirsch, A. Langmann, M. Leitenstern, L. Ögretmen, P. Pitschi, A.-K. Schwehn, C. Schröder, M. Weinmann, F. Werner, B. Lohmann, J. Betz, and M. Lienkamp, "Head-to-head autonomous racing at the limits of handling in the a2rl challenge," 2026, *arXiv preprint arXiv:2602.08571*.
- [17] E. Steiner, D. van der Spuy, F. Zhou, A. Pama, M. Liarokapis, and H. Williams, "Accelerating real-world overtaking in fltenth racing employing reinforcement learning methods," 2025, *arXiv preprint arXiv:2510.26040*.
- [18] S. Tu, X. Zhou, D. Liang, X. Jiang, Y. Zhang, X. Li, and X. Bai, "The role of world models in shaping autonomous driving: A comprehensive survey," 2025, *arXiv preprint arXiv:2502.10498*.
- [19] H. Wang, X. Ye, F. Tao, C. Pan, A. Mallik, B. Yaman, L. Ren, and J. Zhang, "Adawm: Adaptive world model based planning for autonomous driving," in *International Conference on Learning Representations*, 2025.
- [20] P. Yang, B. Lu, Z. Xia, C. Han, Y. Gao, T. Zhang, K. Zhan, X. Lang, Y. Zheng, and Q. Zhang, "Worldrft: Latent world model planning with reinforcement fine-tuning for autonomous driving," in *AAAI Conference on Artificial Intelligence*, 2026.
- [21] G. Wang, P. Tang, X. Ren, G. Zhao, B. Feng, and C. Ma, "Learning vision-language-action world models for autonomous driving," 2026, *arXiv preprint arXiv:2604.09059*.
- [22] J. Shi, J. Chen, Y. Wang, L. Sun, C. Liu, W. Xiong, and T. Wo, "Motion forecasting for autonomous vehicles: A survey," 2025, *arXiv preprint arXiv:2502.08664*.
- [23] S. Ettinger, S. Cheng, B. Caine, C. Liu, H. Zhao, S. Pradhan, Y. Chai, B. Sapp, C. R. Qi, Y. Zhou, Z. Yang, A. Chouard, P. Sun, J. Ngiam, V. Vasudevan, A. McCauley, J. Shlens, and D. Anguelov, "Large scale interactive motion forecasting for autonomous driving: The waymo open motion dataset," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 9710–9719.
- [24] B. Wilson, W. Qi, T. Agarwal, J. Lambert, J. Singh, S. Khandelwal, B. Pan, R. Kumar, A. Hartnett, J. K. Pontes, D. Ramanan, P. Carr, and J. Hays, "Argoverse 2: Next generation datasets for self-driving perception and forecasting," in *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, 2023.
- [25] J. Ngiam, B. Caine, W. Han, B. Yang, Y. Chai, P. Sun, Y. Zhou, X. Yi, O. Alsharif, P. Nguyen, Z. Chen, J. Shlens, and V. Vasudevan, "Scene transformer: A unified architecture for predicting future trajectories of multiple agents," 2021, *arXiv preprint arXiv:2106.08417*.
- [26] N. Nayakanti, R. Al-Rfou, A. Zhou, K. Goel, K. S. Refaat, and B. Sapp, "Wayformer: Motion forecasting via simple and efficient attention networks," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 2980–2989.
- [27] J. Gu, C. Sun, and H. Zhao, "Densetnt: End-to-end trajectory prediction from dense goal sets," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 15 303–15 312.
- [28] Z. Zhou, L. Ye, J. Wang, K. Wu, and K. Lu, "Hivt: Hierarchical vector transformer for multi-agent motion prediction," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 8823–8833.
- [29] Z. Huang, H. Liu, C. L. Wu, and C. Lv, "Gameformer: Game-theoretic modeling and learning of transformer-based interactive prediction and planning for autonomous driving," 2023, *arXiv preprint arXiv:2303.05760*.
- [30] L. Feng, Q. Li, Z. Peng, S. Tan, and B. Zhou, "Trafficgen: Learning to generate diverse and realistic traffic scenarios," in *IEEE International Conference on Robotics and Automation*, 2023.
- [31] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.
- [32] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," in *Proceedings of the International Conference on Machine Learning*, 2018, pp. 1861–1870.
- [33] S. Fujimoto, H. van Hoof, and D. Meger, "Addressing function approximation error in actor-critic methods," in *Proceedings of the International Conference on Machine Learning*, 2018, pp. 1587–1596.